

Sentence Meaning and Participants

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Faculty
of Arts

Roadmap

- 1 Semantic Roles
 - Semantic Roles
- 2 Alternations
- 3 Tense, Aspect and Modality (TAM)



Situations

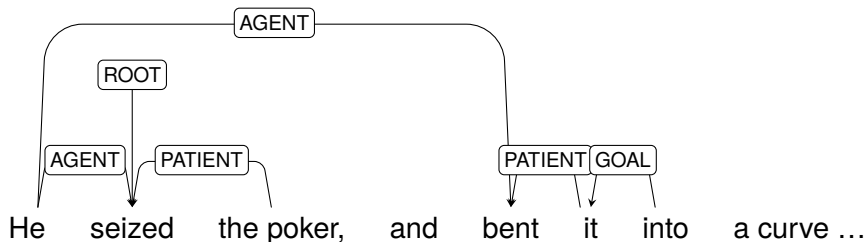
- Noun phrases refer to entities (things, people, places, ...)
- Sentences refer to situations
 - ▶ Situations can be constrained in various ways
 - ▶ What is the event in question?
 - ▶ Who participates in it?
 - ▶ When did it happen?
 - ▶ Is it ongoing or has it finished?
 - ▶ Is our knowledge of it certain?
- The core of an event is typically represented by a verb or adjective
 - ▶ Verbs typically refer to actions (but can refer to states)
He stepped swiftly forward ... DANC
I know you, you scoundrel! DANC
 - ▶ Adjectives typically refer to states
Your sister is dead, then? DANC



Semantic Roles

In this section we talk about the relations between the participants in a situation and the situation itself.

- **Semantic roles** are parts of the sentence that correspond to the participants in the situation described
- They classify relations between entities in a situation

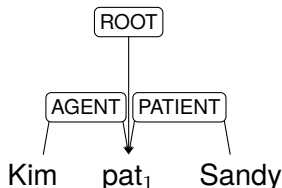


- Also known as
 - ▶ Deep case (Fillmore, 1968)
 - ▶ Thematic roles; Theta roles; θ -roles
 - ▶ Participant Roles

Roles link different alternations

- (1) *Kim patted Sandy*
- (2) *Sandy was patted by Kim*

- The semantic roles are different from the grammatical relations.
- Which is the **subject** and which the **object** in these sentences?
- What are the semantic roles of Kim and Sandy?
- **semantic dependencies**: An abstract representation of the meaning links word-senses to each other using semantic roles: different sentences may end up the same at this level



Semantic Roles I

- **AGENT** (takes *deliberately, on purpose, what did X do?*)
A participant which the meaning of the verb specifies as doing or causing something, possibly intentionally.
 - ▶ The initiator, performer or controller of an action; typically volitional, typically animate
 - ▶ Typically SUBJECT
- (3) Kim kicked Sandy
- (4) The ogre leaped into the fray
- (5) The student watched the video
- (**ACTOR**) generalization of **AGENT** that allows non-volitional, non-actor
if you use this, then **AGENT** is restricted to animate, volitional participants
- Find an example of AGENT in The War with the Newts



Semantic Roles II

- **PATIENT** (*What happened to X?*)

A participant which the verb characterizes as having something happen to it, and as being affected by what happens to it.

- ▶ The undergoer of an action
- ▶ Undergoes change in state usually, both animate and inanimate
- ▶ Typically OBJECT

(6) *Kim kicked Sandy*

(7) *The ogre ate the dog*

(8) *#The student watched the video*

(9) *#I heard a sound*

- Find an example of PATIENT in The War with the Newts



- **THEME**

A participant which is characterized as changing its position or condition, or as being in a state or position.

- ▶ Moved, location or state is described
- ▶ Typically OBJECT

(10) *Hiromi put the book on the shelf*

(11) *Freddy gave you the chocolate*

(12) *The book is on the shelf*

(13) *The protagonist died*

(14) **The dog walked home*

- Find an example of THEME in The War with the Newts



Semantic Roles IV

- **EXPERIENCER**

A participant who is characterized as aware of something.

- ▶ Non-volitional, displaying awareness of action, state
- ▶ Typically SUBJECT

(15) Liling heard thunder

(16) Jo felt sick

(17) The lecturer annoyed the students

- **STIMULUS**

- ▶ Usually used in connection with EXPERIENCER

(18) The lightning scared them

(19) I don't like the lightning

- Find examples of EXPERIENCER and STIMULUS in The War with the Newts

- **BENEFICIARY**

- ▶ For whose benefit the action was performed
- ▶ Typically indexed by **for** PP in English or OBJECT in ditransitive verbs

(20) *They made me a present*

(21) *They made a present for me*

- **LOCATION**

- ▶ Place
- ▶ Typically indexed by locative PPs (**in, on, at**, ...) in English

(22) *I am living in Indonesia*

(23) *It is on the table*

- Find examples of BENEFICIARY and LOCATION in The War with the Newts



Semantic Roles VI

- **GOAL**

- ▶ towards which something moves (lit or metaphor)
- ▶ Typically indexed by **to** PP in English
or OBJECT in ditransitive

(24) *She handed the form to him*

(25) *She handed him her form*

- **SOURCE**

- ▶ from which something moves or originates
- ▶ Typically indexed by
- ▶ from PP in English

(26) *We gleaned this from the Internet*

- Find examples of SOURCE and GOAL in The War with the News?

- **INSTRUMENT**

- ▶ Means by which action is performed
- ▶ Can be indexed by *with* PP in English

(27) *I ate breakfast with chopsticks*

- Find examples of INSTRUMENT in The War with the Newts ?
- Some researchers have a separate COMITATIVE case which encodes a relationship of accompaniment between two participants in an event



PropBank Roles I

- An influential set of semantic roles comes from PropBank (**Palmer et al., 2005**)
- They have 6 core roles and 17 modifier roles
- For the core roles, meaning is defined per verb sense, with typical values given below

Role	Description	Example
ARG0	agent	<i><u>Kim</u>_A <u>walked</u> <u>her dog</u>₀</i>
ARG1	patient	
ARG2	instrument, benefactive, attribute	
ARG3	starting point	
ARG4	ending point	
ARGA	secondary agent	



- **seize.01** “acquire (forcefully or stealthily)”
 - Arg0-PAG: agent, entity acquiring something
 - Arg1-PPT: thing acquired
 - Arg2-DIR: acquired-from

Sandy₀ seized the poker₁ from Kim₂

The Modifier Roles

COM:	Comitative
LOC:	Locative
DIR:	Directional
GOL:	Goal
MNR:	Manner
TMP:	Temporal
EXT:	Extent
REC:	Reciprocals
PRD:	Secondary Predication
PRP:	Purpose
CAU:	Cause
DIS:	Discourse
ADV:	Adverbials
ADJ:	Adjectival
MOD:	Modal
NEG:	Negation
DSP:	Direct Speech
LVB:	Light Verb
CXN:	Construction



Some Issues

- Every theory has a different set of roles
- It is hard to generalize: roles can be very word specific
- Roles are very under-specified, these are all PATIENT!

(28) *The genie touched the lamp with their nose.*

(29) *The baby rubbed the lamp with its hands.*

(30) *The baby squeezed the rubber toy with its hands.*

(31) *She cracked the mirror with a stone.*



Linking Grammatical Relations and Semantic Roles

- Semantic roles typically map onto grammatical functions systematically
 - ▶ AGENT is usually the subject
 - ▶ PATIENT is usually the object
- It is possible to predict how arguments are linked to the verb from their semantic roles, and hence their grammatical functions.
- Many verbs allow **alternation** “syntactic variants with different roles”

(32) *Jo broke the ice with a pickaxe* ⟨AGENT, PATIENT, INSTRUMENT⟩

(33) *The pickaxe broke the ice* ⟨INSTRUMENT, PATIENT⟩

(34) *The ice broke* ⟨PATIENT⟩



Other Predicates

- Adjectives (normally theme)

(35) *John is tall* (THEME)

(36) *John is cold [to touch]* (THEME)

(37) *John is/feels cold* (EXPERIENCER)

different adjectives in e.g., Japanese :

冷たい *tsumetai* “cold (to touch)” vs 寒い *samui* “(feel) cold”

- Predicative Copula (treat second NP as predicate)

(38) *John is a boy* (THEME)

- Identity Copula (reversible)

(39) *Kim is my teacher* (THEME, THEME)?

(40) *My teacher is Kim* (THEME, THEME)?



Roadmap

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- Many verbs have multiple possible mappings of grammatical function to role

- (41) a. *Kim broke the window with the hammer*
b. *The hammer broke the window*
c. *The window broke*

- (42) a. *I cut the cake with the knife*
b. *This cake cuts easily*

- The relations between them are called **alternations**
- English Verb Classes and Alternations* (Levin, 1993)

There are many alternations I

- A common way to change the number of arguments is called **voice**: passive, middle

(43) **Transitive Passive**

- Kim ate Sandy*
- Sandy was eaten (by Kim)*

(44) **Ditransitive Passive**

- Abraham gave Brown chocolate*
- Abraham gave chocolate to Brown*
- Chocolate was given to Brown (by Abraham)*
- Brown was given chocolate (by Abraham)*



There are many alternations II

(45) **Transitive Middle** (or just causative/inchoative)

- a. *They open the gate very quietly*
- b. *The gate opens very quietly*

(46) **Intransitive Middle**

- a. *The knife cuts the cake well*
- b. *The knife cuts well*

- And some more alternations:

(47) **Conative** alternation:

- a. *Kim hit the door* ↔ *Kim hit at the door*

(48) **Body-part possessor ascension** alternation:

- a. *Kim cut Sandy's arm* ↔ *Kim cut Sandy on the arm*



Why so many possibilities?

- So we can emphasize different participants
- We may not know all the participants
- We may not care about all the participants
- There are also lexical alternations

(49) *Kim killed Sandy* vs *Sandy dies*

(50) c.f. *Kim melted the ice* vs *the ice melted*

(51) 金が 氷を 溶かした vs 氷が 溶けた
Kim-ga koori-wo tokashita koori-ga toketa
Kim-SBJ ice-OBJ *melt:trans* ice-SBJ *melt:intrans*



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Tense, Aspect and Modality

- We need to distinguish grammatical expression from meaning
 - ▶ Tense vs Time
 - ▶ Grammatical Aspect vs Semantic Aspect
 - ▶ Mood vs Modality
 - ▶ Surface Case vs Deep Case
- The relation between them is referred to as
 - ▶ linking; syntax-semantics interface; grammar



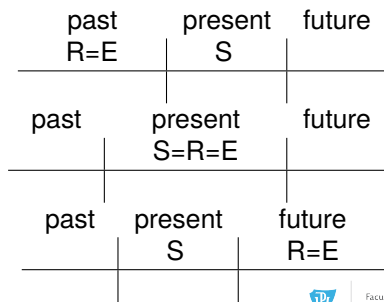
How Universal is Tense?

- Grammatical tense is different from semantic time
- English has **past/non-past**
- Latin marks **past/present/future**
- Chibemba (Bantu) has **metrical tense**
 - ▶ Remote Past (< yesterday)
 - ▶ Removed Past (yesterday)
 - ▶ Near Past (today)
 - ▶ Immediate Past (past few hours)
 - ▶ Immediate Future (next few hours)
 - ▶ Near Future (today)
 - ▶ Removed Future (tomorrow)
 - ▶ Remote Future (> tomorrow)



Tense and Time

- Locate a situation with respect to a point in time
 - ▶ S = speech point
 - ▶ R = reference time
 - ▶ E = event time
- Hans Reichenbach (1947)
- Simple Tense
 - ▶ Past ($R = E < S$) *saw*
 - ▶ Present ($R = S = E$) *see*
 - ▶ Future ($S < R = E$) *will see*



Complex Tense

- Past Perfect ($E < R < S$) *had seen*

past		present	future
E	R	S	

By 1939 my Father had seen many arrests

- Future Perfect ($S < E < R$) *will have seen*

past	present	future
	S	E R

By 2039 my son will have seen many things



- Finer grained talking about time!
- **Progressive** is used for ongoing processes (unfinished)
 - ▶ **Past Progressive** *I was building the building*
 - ▶ **Present Progressive** *I am building the building*
 - ▶ **Future Progressive** *I will be building the building*
- **Perfect** compares the time to the reference point
 - ▶ **Past Perfect** *I had built the building* ($E < R < S$)
 - ▶ **Present Perfect** *I have built the building* ($E < R = S$)
 - ▶ **Future Perfect** *I will have built the building* ($S < E < R$)

Aspect more Generally

- **Perfective** focuses on the end point
 - ▶ **Completive** *I built the building*
 - ▶ **Experiential** *I have built the building*
- **Imperfective**
 - ▶ **Progressive** *I was listening/I am listening*
 - ▶ **Habitual** *I listen to the Goon Show*
- Different languages grammaticalize different things



Mood and Modality

- Modality expresses varying degrees of the speaker's commitment and belief
- In English it is typically expressed by an auxiliary verb.

(52) *She has left by now.*

(53) *She must have left by now.*

(54) *She could have left by now.*

(55) *She needn't have left by now.*

(56) *She couldn't have left by now.*

(57) *She has to leave by now.*

(58) *She must leave by now.*

(59) *She can leave now.*



Other means of expression

- Explicit External Verb

(60) *I know that S*

(61) *I believe that S*

- Adverb or Adjective

(62) *It is certain that S*

(63) *It is likely that S*

(64) *I will probably S*

(65) *I will definitely S*



Knowledge vs Obligation

- **Epistemic modality**: Speaker signals degree of knowledge.

(66) *You can drive this car* (You have the ability to)

- **Deontic modality**: Speaker signals his/her attitude to social factors of obligation and permission.

- ▶ **Permission**

(67) *You can drive this car* (You have permission to)

(68) *You may drive this car*

- ▶ **Obligation**

(69) *You must drive this car* (You have an obligation to)

(70) *You ought to drive this car*



Other modalities

There are other possible modalities (Kroeger, 2022, p290 (4)):

- **bouletic** concerns what is possible or necessary, given a person's desires
- **dynamic** concerns properties and dispositions of persons (such as ability)
- **teleological** concerns achieving a goal

(71) *It has to be raining.* [after observing people coming inside with wet umbrellas] **epistemic**

(72) *Visitors have to leave by six pm.* [hospital regulations] **deontic**

(73) *John has to work hard if he wants to retire at age 50.* [to attain desires] **bouletic**

(74) *I have to sneeze.* [given the current state of one's nose] **dynamic**

(75) *To get home in time, you have to take a taxi.* [in order to achieve the stated purpose] **teleological**

Possible Worlds

- We can analyze these in terms of **possible worlds**
- We mark how close a hypothetical case is to reality:

(76) *It must be/might be/is/can't be hot outside*

And model it as the degree of overlap of the worlds

- Similarly for **conditionals** (condition/consequence)

(77) *If it is Singapore, it will be hot outside*

(78) *If it were Singapore, it would be hot outside*

(79) *If you should go to Singapore, take some cool clothes*



Mood more Generally

- Grammatical Inflection used to mark modality is called **mood**
 - indicative** expresses factual statements
 - conditional** expresses events dependent on a condition
 - imperative** expresses commands
 - injunctive** expresses pleading, insistence, imploring
 - optative** expresses hopes, wishes or commands
 - potential** expresses something likely to happen
 - subjunctive** expresses hypothetical events; opinions or emotions
 - interrogative** expresses questions

- In English most things are marked syntactically:

(80)	<i>I <u>am</u> good</i>	indicative
(81)	<i><u>Am</u> I good?</i>	interrogative
(82)	<i><u>Be</u> good!</i>	imperative
(83)	<i>If I <u>were</u> a rich man</i>	subjunctive

- ? Find examples of non-indicative sentences in The War with the
Newts

Evidentiality

Some languages must show how you gained the evidence

- **nonvisual sensory**: speaker felt the sensation
 - ▶ /p^ha · bék^h-ink'e/ “burned, I felt it”
- **inferential**: speaker saw circumstantial evidence
 - ▶ /p^ha · bék-ine/ “must have burned”
- **hearsay (reportative)**: speaker is reporting what was told
 - ▶ /p^ha · bék^h-le/ “burned, they say”
- **direct knowledge**: speaker has direct evidence, probably visual
 - ▶ /p^ha · bék-a/ “burned, I saw it”
- Examples from Eastern Pomo (McLendon 2003)



Evidentiality in English

We can, and often do, mark evidentiality in English, although it is not strongly grammaticalized.

- (84) *Bob is hungry.*
- (85) *Bob looks hungry.*
- (86) *Bob seems hungry.*
- (87) *Bob is apparently hungry.*
- (88) *Bob would be hungry by now.*
- (89) *Look at those clouds! It's going to rain!*
- (90) *Look at those clouds! # It will rain!.*



Conclusions

- Situations are represented by verbs
- Semantic roles can be used to make the relations between the situation and the participants clearer
- We can say something about the situation, in particular Tense, Aspect and Modality (TAM)
- **Saeed (2009)** talks about Tense, Aspect, Modality and Evidentiality in Chapter 5 and Semantic Roles (Thematic Roles) in Chapter 6
- **Kroeger (2022)** talks about Modals and Evidentiality in Part V, and Tense and Aspect in part VI



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Glossary of Key Terms (English–Czech) I

English	Čeština
analysis	analýza
autonomous	autonomní
actor	aktor
agent	agens
alternation(s)	alternace
ambiguous	víceznačný
beneficiary	beneficient
body-part possessor ascension	vzestup posesora části těla
completive	kompletivní
compositional semantics	kompoziční sémantika
conative	konativní
conditional(s)	kondicionál
construction(s)	konstrukce
content word	lexikální slovo



Glossary of Key Terms (English–Czech) II

deontic modality	deontická modalita
differentiae	rozlišující rysy
direct knowledge	přímá znalost
ditransitive passive	ditranzitivní pasivum
epistemic modality	epistemická modalita
experiencer	proživatel
experiential	experenciální
function word	funkční slovo
future perfect	předbudoucí čas
future progressive	budoucí průběhový čas
genus	rod (genus)
goal	cíl
habitual	habituální
hearsay (reportative)	nepřímá řeč (reportativ)
homonyms	homonyma
hyponyms	hyponymie



Glossary of Key Terms (English–Czech) III

imperative	imperativ
imperfective	nedokonavý
indicative	indikativ
inferential	inferenciální
injunctive	injunktiv
instrument	instrument
interpretation	interpretace
interrogative	tázací
intransitive middle	intransitivní medium
irrealis	irrealis
lexeme	lexém
location	lokace
manner	způsob
metrical tense	metrický čas
mood	slovesný způsob
multi-word expressions	víceslovné výrazy



Glossary of Key Terms (English–Czech) IV

nonvisual sensory	mimovizuální smyslový
object	předmět (objekt)
obligation	povinnost
optative	optativ
past perfect	předminulý čas
past progressive	minulý průběhový čas
patient	patient
perfect	perfektum
perfective	dokonavý
permission	dovolení
polysemous	mnohoznačný
possible worlds	možné světy
potential	potenciál
present perfect	předpřítomný čas
present progressive	přítomný průběhový čas
progressive	průběhový



Glossary of Key Terms (English–Czech) V

projection rules

realis

restriction

selectional preferences / restrictions

semantic dependencies

semantic roles

senses and relations

signs

source

state

stimulus

subject

subjunctive

theme

time

transitive middle

projekční pravidla

realis

restrikce

selekční preference / restrikce

sémantické závislosti

sémantické role

významy a vztahy

znaky

zdroj

stav

stimul

podmět (subjekt)

konjunktiv

téma

čas

tranzitivní medium



Glossary of Key Terms (English–Czech) VI

transitive passive

vagueness

voice

word

tranzitivní pasivum

vágnost

slovesný rod

slovo

